Chapter 2: Basic Topology

Section: Compact Sets

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Definitions

Compactness Is Intrinsic

Compactness and Closedness

The Finite Intersection Property

Nested Intervals and k-Cells

k-Cells Are Compact

Heine-Borel and Weierstrass

Why Compactness Matters

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- Compactness is a finiteness condition on infinite sets
- It generalizes “closed and bounded” from $\mathbb{R}$ to metric spaces
- It is the right hypothesis for continuity theorems (Chap. 4)

Definitions

Definition 2.31: Open Cover

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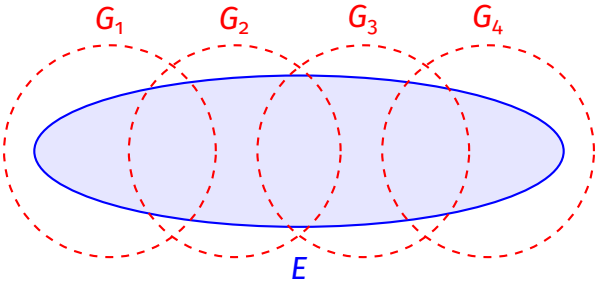
An open cover of a set E in a metric space X is a collection $\{G_\alpha\}$ of open subsets of X such that

$$E \subset \bigcup_\alpha G_\alpha.$$

Key idea: The open sets G_α together "cover" every point of E.

Visualizing an Open Cover



$$E \subset G_1 \cup G_2 \cup G_3 \cup G_4$$

Definition 2.32: Compact Set

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A subset K of a metric space X is compact if every open cover of K contains a finite subcover.

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A subset K of a metric space X is compact if every open cover of K contains a finite subcover.

More explicitly: if $\{G_\alpha\}$ is an open cover of K, then there are finitely many indices $\alpha_1, \ldots, \alpha_n$ such that

$$K \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}.$$

Remarks on Compactness

Remarks

- Every *finite* set is trivially compact.
- Infinite compact sets in $\mathbb{R}^k$ exist (Theorem 2.41).
- Compactness is fundamental for continuity (Chap. 4).

Compactness Is Intrinsic

Theorem 2.33: Compactness Is Intrinsic

Theorem 2.33

Suppose $K \subset Y \subset X$. Then K is compact relative to X if and only if K is compact relative to Y.

Key idea: Unlike "open" or "closed," compactness does not depend on the ambient metric space.

Proof Strategy: Theorem 2.33

Goal: Show compactness relative to $X \iff$ compactness relative to Y.

Key tool: Theorem 2.30 — open sets of Y are exactly $V_\alpha = Y \cap G_\alpha$ where G_α is open in X.

Approach:

- ($\Rightarrow$) Given a Y-open cover, lift to an X-open cover.
- ($\Leftarrow$) Given an X-open cover, restrict to Y.

Proof of Theorem 2.33 — Part 1: X compact $\Rightarrow$ Y compact

Step 1: Lift the cover

Let $\{V_\alpha\}$ be open in Y with $K \subset \bigcup_\alpha V_\alpha$. By Theorem 2.30, there exist G_α open in X such that

$$V_\alpha = Y \cap G_\alpha.$$

Proof of Theorem 2.33 — Part 1: X compact $\Rightarrow$ Y compact

Step 2: Apply X-compactness

Since $K \subset Y \subset \bigcup_\alpha G_\alpha$ and K is compact in X:

$$K \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}. \tag{22}$$

Since $K \subset Y$, intersecting with Y gives:

$$K \subset V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}. \tag{23}$$

Proof of Theorem 2.33 — Part 2: Y compact $\Rightarrow$ X compact

Reverse direction

Let $\{G_\alpha\}$ be open in X covering K. Put $V_\alpha = Y \cap G_\alpha$.

- Each V_α is open in Y, and $\{V_\alpha\}$ covers K.
- By Y-compactness: $K \subset V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}$.
- Since $V_\alpha \subset G_\alpha$: $K \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}$. $\square$

Compactness and Closedness

Theorem 2.34: Compact $\Rightarrow$ Closed

Theorem 2.34

Compact subsets of metric spaces are closed.

Strategy: Show the complement K^c is open by proving that every point outside K is an interior point of K^c.

Proof of Theorem 2.34 — Setup

Setup

Let K be compact and fix $p \in X$, $p \notin K$.

For each $q \in K$, let V_q, W_q be neighborhoods of p, q of radius less than $\frac{1}{2}d(p, q)$.

- $V_q \cap W_q = \varnothing$ (disjoint, by construction).
- $\{W_q\}_{q \in K}$ is an open cover of K.

Proof of Theorem 2.34 — Finish

Apply compactness

By compactness, finitely many points $q_1, \ldots, q_n \in K$ satisfy:

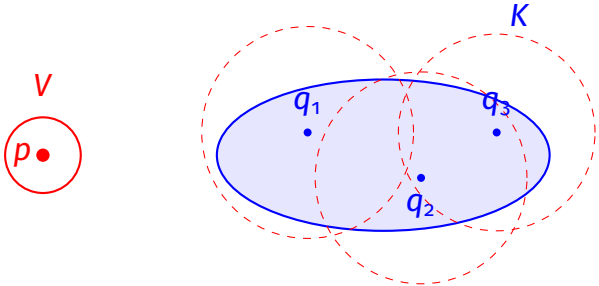
$$K \subset W_{q_1} \cup \cdots \cup W_{q_n} = W.$$

Let $V = V_{q_1} \cap \cdots \cap V_{q_n}$.

- V is a neighborhood of p (finite intersection).
- $V \cap W = \varnothing$, so $V \subset K^c$.

Hence p is interior to K^c. $\square$

Visualizing the Separation



$V = V_{q_1} \cap V_{q_2} \cap V_{q_3}$ is disjoint from $W = W_{q_1} \cup W_{q_2} \cup W_{q_3}$

Theorem 2.35: Closed Subsets of Compact Sets Are Compact

Theorem 2.35

Closed subsets of compact sets are compact.

Strategy: Given an open cover of F, add F^c to get an open cover of K, extract a finite subcover, and discard F^c.

Proof of Theorem 2.35

Proof

Let $F \subset K$, F closed, K compact. Let $\{V_\alpha\}$ be an open cover of F.

- $\Omega = \{V_\alpha\} \cup \{F^c\}$ is an open cover of K (since F^c is open).
- K compact $\Rightarrow$ finite subcollection $\Phi \subset \Omega$ covers K.
- If $F^c \in \Phi$, remove it. What remains still covers F. $\square$

Corollary: Intersection of Closed and Compact

Corollary to Theorem 2.35

If F is closed and K is compact, then $F \cap K$ is compact.

Proof

- K is closed by Theorem 2.34.
- $F \cap K$ is closed by Theorem 2.24(b) (intersection of closed).
- $F \cap K \subset K$, so by Theorem 2.35, it is compact. $\square$

The Finite Intersection Property

Theorem 2.36: Finite Intersection Property

Theorem 2.36

Let $\{K_\alpha\}$ be compact subsets of a metric space X. If every finite subcollection has nonempty intersection, then

$$\bigcap_\alpha K_\alpha \neq \varnothing.$$

Key idea: Testing intersection finitely at a time is enough.

Proof of Theorem 2.36 — Step 1: Setup (by contradiction)

Suppose $\bigcap K_\alpha = \varnothing$

Fix a member K_1 of the collection. Put $G_\alpha = K_\alpha^c$.

Assume no point of K_1 belongs to every K_α:

$$K_1 \cap \bigcap_\alpha K_\alpha = \varnothing \iff K_1 \subset \bigcup_\alpha G_\alpha.$$

Proof of Theorem 2.36 — Step 2: Finite Subcover

Apply compactness of K_1

By compactness of K_1, there exist $\alpha_1, \ldots, \alpha_n$ with

$$K_1 \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}.$$

Taking complements:

$$K_1 \cap K_{\alpha_1} \cap \cdots \cap K_{\alpha_n} = \varnothing.$$

But this contradicts the finite intersection hypothesis! $\square$

Corollary: Nested Compact Sets

Corollary to Theorem 2.36

If $\{K_n\}$ is a sequence of nonempty compact sets with

$$K_1 \supset K_2 \supset K_3 \supset \cdots,$$

then $\bigcap_{n=1}^{\infty} K_n \neq \varnothing$.

Reason: Any finite subcollection is nested, so its intersection equals the smallest member $K_N \neq \varnothing$.

Theorem 2.37: Infinite Subsets of Compact Sets

Theorem 2.37

If E is an infinite subset of a compact set K, then E has a limit point in K.

Contrapositive strategy: If no point of K is a limit point of E, produce an open cover with no finite subcover.

Proof of Theorem 2.37

Proof by contradiction

Assume no $q \in K$ is a limit point of E.

- Each $q \in K$ has a neighborhood V_q with $|V_q \cap E| \leq 1$.
- $\{V_q\}$ is an open cover of K.
- No finite subcollection can cover E (covers at most finitely many points).
- Since $E \subset K$, no finite subcollection covers K.
- This contradicts compactness. $\square$

Nested Intervals and k-Cells

Theorem 2.38: Nested Intervals in $\mathbb{R}^1$

Theorem 2.38

If $\{I_n\}$ is a sequence of closed intervals in $\mathbb{R}^1$ with $I_1 \supset I_2 \supset \cdots$, then

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Tool: The least upper bound property of $\mathbb{R}$.

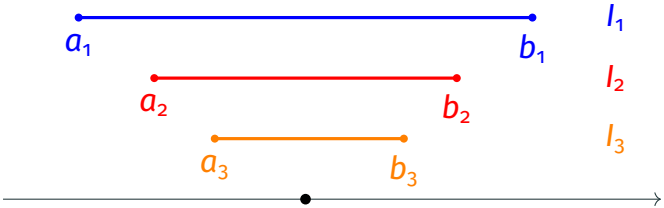
Proof of Theorem 2.38

Proof

Let $I_n = [a_n, b_n]$. Put $E = \{a_n : n \geq 1\}$.

- E is nonempty and bounded above by b_1.
- Let $x = \sup E$.
- For any m, n: $a_n \leq a_{m+n} \leq b_{m+n} \leq b_m$.
- So $x \leq b_m$ for every m, and $a_m \leq x$ for every m.
- Hence $x \in [a_m, b_m] = I_m$ for every m. $\square$

Visualizing Nested Intervals



$$x = \sup\{a_n\}$$

Theorem 2.39: Nested k-Cells

Theorem 2.39

If $\{I_n\}$ is a sequence of k-cells with $I_1 \supset I_2 \supset \cdots$, then $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$.

Recall: a *k-cell* is a product $[a_1, b_1] \times \cdots \times [a_k, b_k]$.

Strategy: Apply Theorem 2.38 coordinate-by-coordinate.

Proof of Theorem 2.39

Proof

Let I_n consist of $\mathbf{x} = (x_1, \ldots, x_k)$ with $a_{n,j} \leq x_j \leq b_{n,j}$.

Put $I_{n,j} = [a_{n,j}, b_{n,j}]$. For each fixed j:

- $\{I_{n,j}\}_n$ is a nested sequence in $\mathbb{R}^1$.
- By Theorem 2.38, some $x_j^* \in I_{n,j}$ for all n.

Setting $\mathbf{x}^* = (x_1^*, \ldots, x_k^*)$: $\mathbf{x}^* \in I_n$ for all n. $\square$

k-Cells Are Compact

Theorem 2.40: Every k-Cell Is Compact

Theorem 2.40

Every k-cell is compact.

Technique: Proof by contradiction — bisection.

- Suppose some open cover has no finite subcover.
- Repeatedly halve the cell; one half still fails.
- The nested halves shrink to a point, which must lie in a single cover element — contradiction.

Proof of Theorem 2.40 — Setup

Setup

Let I be a k-cell with $a_j \leq x_j \leq b_j$. Define the diameter

$$\delta = \left\{ \sum_{j=1}^{k} (b_j - a_j)^2 \right\}^{1/2}.$$

Then $\mathbf{x}, \mathbf{y} \in I \Rightarrow |\mathbf{x} - \mathbf{y}| \leq \delta$.

Assume: $\{G_\alpha\}$ is an open cover of I with no finite subcover.

Proof of Theorem 2.40 — Step 1: Bisection

Halve each coordinate

Let $c_j = (a_j + b_j)/2$. The intervals $[a_j, c_j]$ and $[c_j, b_j]$ determine 2^k subcells $Q_1, \ldots, Q_{2^k}$ with $\bigcup Q_i = I$.

If every Q_i had a finite subcover, so would I. So at least one Q_i, call it I_1, has no finite subcover.

Proof of Theorem 2.40 — Step 2: Iterate

Continue bisecting

Subdivide I_1 and keep a failing half. Iterate. This yields $\{I_n\}$:

(a) $I \supset I_1 \supset I_2 \supset \cdots$

(b) I_n has no finite subcover from $\{G_\alpha\}$.

(c) $\mathbf{x}, \mathbf{y} \in I_n \Rightarrow |\mathbf{x} - \mathbf{y}| \leq 2^{-n}\delta$.

Proof of Theorem 2.40 — Step 3: The Contradiction

Reach the contradiction

By (a) and Theorem 2.39, some $\mathbf{x}^* \in \bigcap_n I_n$.

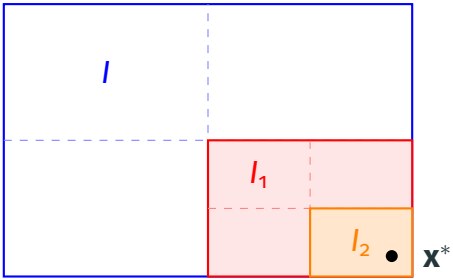
Pick α with $\mathbf{x}^* \in G_\alpha$. Since G_α is open, $\exists\, r > 0$: $|\mathbf{y} - \mathbf{x}^*| < r \Rightarrow \mathbf{y} \in G_\alpha$.

Choose n so large that $2^{-n}\delta < r$. By (c),

$$I_n \subset G_\alpha,$$

a finite (single-element) subcover — contradicting (b). $\square$

Visualizing the Bisection Argument (for $k = 2$)



Heine-Borel and Weierstrass

Theorem 2.41: Heine-Borel

Theorem 2.41 (Heine-Borel)

For a set $E \subset \mathbb{R}^k$, the following are equivalent:

(a) E is closed and bounded.

(b) E is compact.

(c) Every infinite subset of E has a limit point in E.

Note: The equivalence of (a) and (b) is the classical Heine-Borel theorem.

Proof Strategy: Theorem 2.41

Cyclic proof: (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (a).

- (a) $\Rightarrow$ (b): closed + bounded $\subset$ k-cell $\Rightarrow$ closed subset of compact, by Thm 2.40 & 2.35.
- (b) $\Rightarrow$ (c): direct consequence of Theorem 2.37.
- (c) $\Rightarrow$ (a): if not bounded or not closed, construct an infinite set with no limit point in E.

Proof — (a) $\Rightarrow$ (b)

Closed + bounded $\Rightarrow$ compact

- E bounded $\Rightarrow$ $E \subset I$ for some k-cell I.
- I is compact (Theorem 2.40).
- E is closed, so E is a closed subset of the compact set I.
- By Theorem 2.35, E is compact. $\square$

Proof — (b) $\Rightarrow$ (c)

Compact $\Rightarrow$ every infinite subset has a limit point in E

This is exactly Theorem 2.37. $\square$

Proof — (c) $\Rightarrow$ (a): E is bounded

If E is unbounded, (c) fails

Suppose E is not bounded. Choose $\mathbf{x}_n \in E$ with $|\mathbf{x}_n| > n$.

- $S = \{\mathbf{x}_n\}$ is infinite.
- Any $\mathbf{y} \in \mathbb{R}^k$: only finitely many $\mathbf{x}_n$ satisfy $|\mathbf{x}_n - \mathbf{y}| < 1$ (since $|\mathbf{x}_n| > n$).
- So S has no limit point in $\mathbb{R}^k$, hence none in E.
- Contradicts (c). Therefore E is bounded.

Proof — (c) $\Rightarrow$ (a): E is closed (setup)

Suppose E is not closed

Let $\mathbf{x}_0$ be a limit point of E with $\mathbf{x}_0 \notin E$.

For each n, choose $\mathbf{x}_n \in E$ with $|\mathbf{x}_n - \mathbf{x}_0| < 1/n$. Let $S = \{\mathbf{x}_n\}$.

- S is infinite (otherwise $|\mathbf{x}_n - \mathbf{x}_0|$ has a positive lower bound, contradiction).
- $\mathbf{x}_0$ is a limit point of S.

Proof — (c) $\Rightarrow$ (a): E is closed (finish)

S has no other limit point in $\mathbb{R}^k$

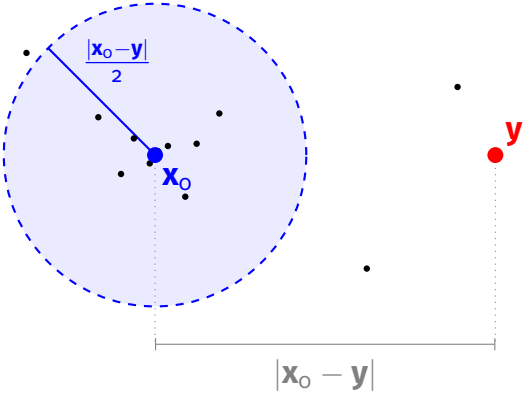
For $\mathbf{y} \neq \mathbf{x}_0$, by the triangle inequality:

$$|\mathbf{x}_n - \mathbf{y}| \geq |\mathbf{x}_0 - \mathbf{y}| - \tfrac{1}{n} \geq \tfrac{1}{2}|\mathbf{x}_0 - \mathbf{y}|$$

for large n. So $\mathbf{y}$ is not a limit point of S (Thm 2.20).

$S \subset E$, infinite, but its only limit point $\mathbf{x}_0 \notin E$. This contradicts (c). $\square$

Visualizing the Closure Argument



All but finitely many $\mathbf{x}_n$ lie inside the ball — so $\mathbf{y}$ is not a limit point of S.

Remark on Theorem 2.41

Scope of Heine-Borel

- (b) $\iff$ (c) holds in *any* metric space (Exercise 26).
- (a) $\Rightarrow$ (b) is special to $\mathbb{R}^k$.
- Counterexamples: $\mathscr{L}^2$ (Chap. 11), Exercise 16.

Theorem 2.42 (Weierstrass)

Theorem 2.42 (Bolzano-Weierstrass)

Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.

Proof

- E bounded $\Rightarrow E \subset I$ for some k-cell I.
- I compact (Theorem 2.40).
- E infinite subset of compact $I \Rightarrow E$ has a limit point in I (Theorem 2.37). $\square$

Summary

Key takeaways from this section:

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Key takeaways from this section:

- A set is compact if every open cover has a finite subcover (Def. 2.32).
- Compactness is *intrinsic*: independent of ambient space (Thm. 2.33). Compact sets are closed (Thm. 2.34), and closed subsets of compact sets are compact (Thm. 2.35).
- Finite intersection property: a collection of compact sets with nonempty finite intersections has nonempty total intersection (Thm. 2.36). Infinite subsets of compact sets have limit points in the set (Thm. 2.37).

Summary

Key takeaways from this section:

- Every k-cell is compact (Thm. 2.40) via bisection and nested k-cells (Thms. 2.38, 2.39). Heine-Borel (Thm. 2.41): in $\mathbb{R}^k$, closed and bounded $\iff$ compact $\iff$ every infinite subset has a limit point in the set.

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Key takeaways from this section:

- Every k-cell is compact (Thm. 2.40) via bisection and nested k-cells (Thms. 2.38, 2.39). Heine-Borel (Thm. 2.41): in $\mathbb{R}^k$, closed and bounded $\iff$ compact $\iff$ every infinite subset has a limit point in the set.
- Bolzano-Weierstrass (Thm. 2.42): every bounded infinite subset of $\mathbb{R}^k$ has a limit point.

Coming up next: Perfect sets and the Cantor set.